PHARMACEUTICAL DEVIATION INCIDENT REPORT

Quality Assurance Department | GMP Manufacturing Operations | Form QA-DEV-04

1. INCIDENT IDENTIFICATION & CLASSIFICATION

Deviation Number:	DEV-2026-019
Title:	Out-of-Specification (OOS) Dissolution Failure in Metformin HCl 500mg Tablets
Date of Occurrence:	2026-10-14
Date Reported:	2026-10-15
Reported By:	Sarah Jenkins (QC Senior Analyst)
Facility / Site:	Building 4, Solid Oral Dosage Manufacturing Facility
Severity Classification:	Critical Deviation (Potential Patient Efficacy Impact)

2. PRODUCT & BATCH TRACEABILITY

Product Name:	Metformin Hydrochloride Extended-Release Tablets USP 500mg
Batch / Lot Number:	MET-2026-B849
Manufacturing Date:	2026-10-10
Expiry Date:	2028-10-09
Batch Size:	450,000 Tablets (Drum lots 01 through 09)
Equipment Involved:	Fette 2090 Rotary Tablet Press (TP-04), Glatt Fluid Bed Granulator (FBG-02)

3. DESCRIPTION OF DEVIATION

During finished product release testing of batch MET-2026-B849, Stage 1 (S1) dissolution testing yielded a mean result of 68.4% release at Q=45 minutes against the USP monograph specification of NLT 80% (Q). Two individual units tested below 65% (Unit #3: 63.1%, Unit #5: 64.8%). Re-testing in Stage 2 (S2) confirmed the failure with an overall 12-unit mean of 71.2%.

Immediate Containment Actions: Quarantine Hold Notice QHN-2026-118 was issued. All 9 drums of lot MET-2026-B849 were physically locked in Quarantine Warehouse Bay C. No product was released to market.

4. PRELIMINARY INVESTIGATION & OBSERVATIONS

- Raw Materials: Metformin API batch API-M-902 passed all chemical and particle size (d90=42um) specifications.
- Granulation: Fluid bed granulation drying temperature logged at 58C, within the 55-62C target range.
- Compression Stage: Tablet compression log for Fette 2090 indicates main compression force was operated at 28.5 kN, which significantly exceeds the validated master batch record range of 18.0 - 22.0 kN. Tablet breaking force (hardness) was recorded at 14.8 kp (specification: 7.0 - 10.5 kp).
- Tooling & Operator: Senior compression operator was on emergency leave; replacement relief operator adjusted compression force manually after observing slight tablet capping during initial startup without QA supervisor sign-off.

5. SCOPE & REGULATORY IMPACT

- Impact Assessment: Batch is rejected and slated for destruction. Sister batch MET-2026-B850 manufactured on the same shift has been quarantined pending validation check.
- Regulatory Reporting: Triggered 3-day notification requirement under 21 CFR 211.192 and EU GMP Chapter 1.
- Investigation Target Date: 2026-10-28 (Within standard 30-day deviation closure window).